

Linux Commands Assignment

1. Creating and Renaming Files/Directories

Create a Directory

Command:

```
mkdir test_dir
```

Explanation:

Creates a new directory named `test_dir`.

Create an Empty File

Command:

```
touch test_dir/example.txt
```

Explanation:

Creates an empty file named `example.txt` inside the `test_dir` directory.

Rename the File

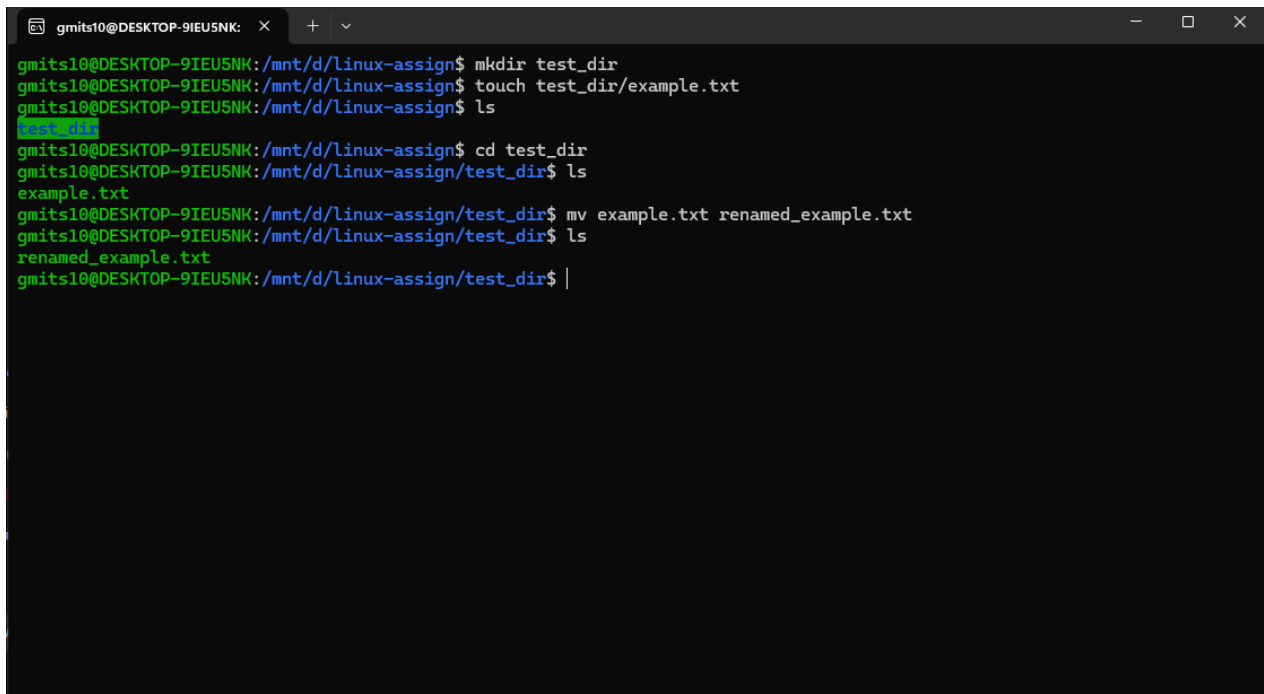
Command:

```
mv test_dir/example.txt test_dir/renamed_example.txt
```

Explanation:

Renames `example.txt` to `renamed_example.txt`.

Screenshot:

A terminal window with a dark background and light green text. The window title is 'gmits10@DESKTOP-9IEU5NK:'. The commands and their outputs are as follows:

```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ mkdir test_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ touch test_dir/example.txt
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ ls
test_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ cd test_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/test_dir$ ls
example.txt
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/test_dir$ mv example.txt renamed_example.txt
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/test_dir$ ls
renamed_example.txt
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/test_dir$ |
```

2. Viewing File Contents

Display Entire File

Command:

```
cat /etc/passwd
```

Explanation:

Displays the complete contents of the `/etc/passwd` file.

Display First 5 Lines

Command:

```
head -n 5 /etc/passwd
```

Explanation:

Shows only the first five lines of the file.

Display Last 5 Lines

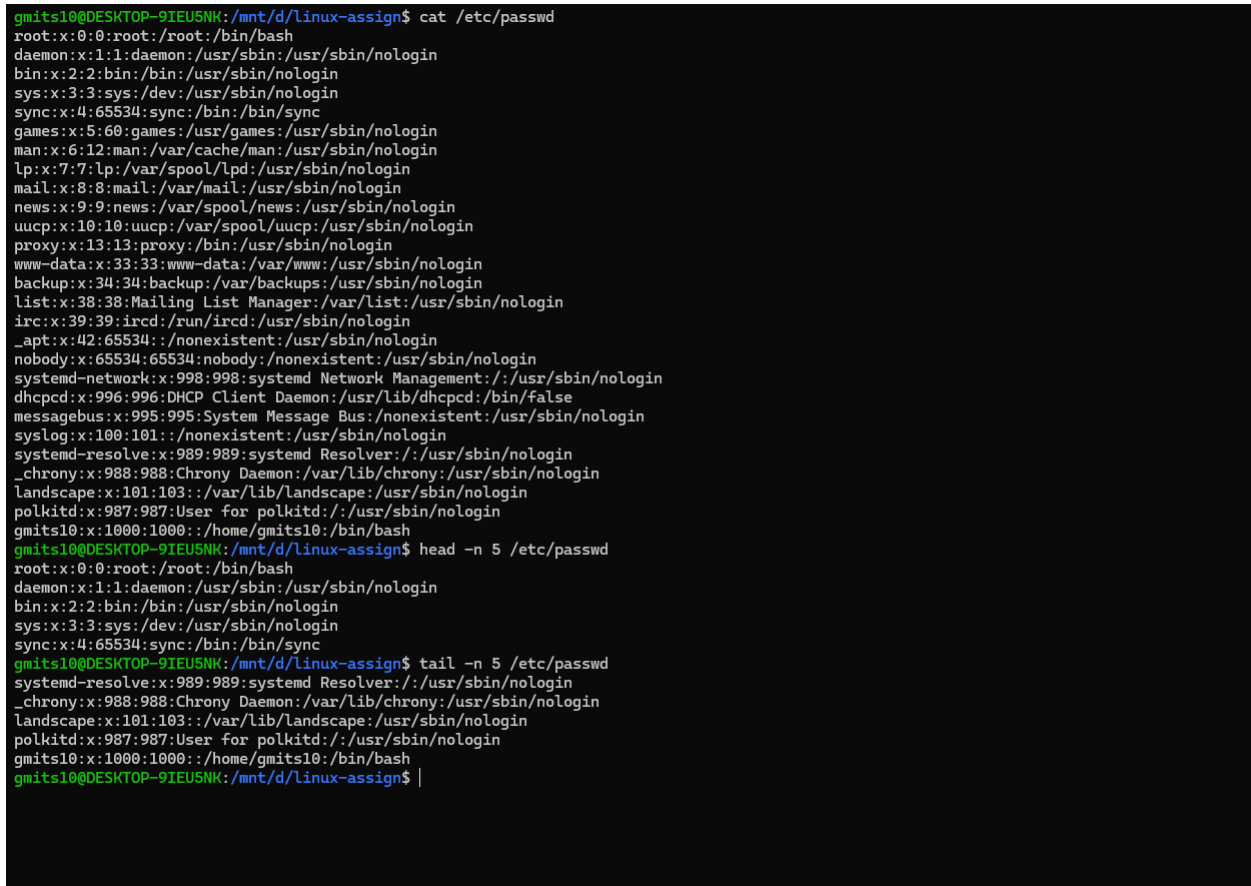
Command:

```
tail -n 5 /etc/passwd
```

Explanation:

Shows only the last five lines of the file.

Screenshot:



```
gmits10@DESKTOP-9IEUSNK:/mnt/d/linux-assign$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:./usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:./usr/sbin/nologin
gmits10:x:1000:1000:./home/gmits10:/bin/bash
gmits10@DESKTOP-9IEUSNK:/mnt/d/linux-assign$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
gmits10@DESKTOP-9IEUSNK:/mnt/d/linux-assign$ tail -n 5 /etc/passwd
systemd-resolve:x:989:989:systemd Resolver:./usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:./usr/sbin/nologin
gmits10:x:1000:1000:./home/gmits10:/bin/bash
gmits10@DESKTOP-9IEUSNK:/mnt/d/linux-assign$ |
```

3. Searching for Patterns

Find Lines Containing "root"

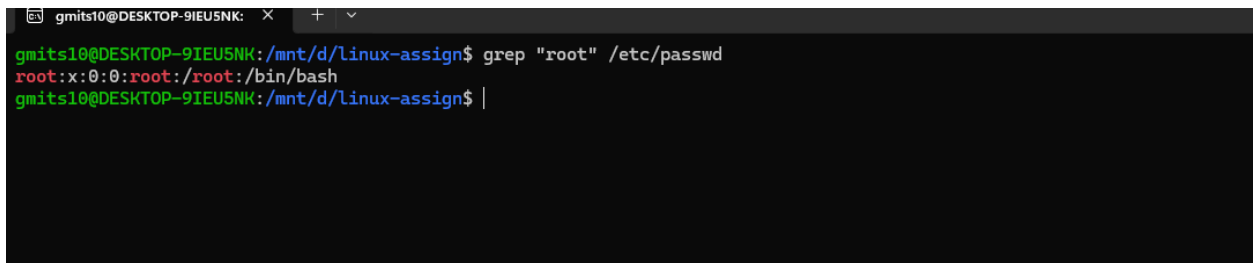
Command:

```
grep "root" /etc/passwd
```

Explanation:

Searches for and displays all lines containing the word `root`.

Screenshot:

A screenshot of a terminal window with a dark background. The window title is 'gmits10@DESKTOP-9IEU5NK:'. The prompt is 'gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign\$'. The command 'grep "root" /etc/passwd' has been entered. The output shows a single line: 'root:x:0:0:root:/root:/bin/bash'. The prompt then changes to 'gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign\$' with a cursor at the end.

```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ |
```

4. Zipping and Unzipping

Compress Directory

Command:

```
zip -r test_dir.zip test_dir
```

Explanation:

Creates a ZIP archive named `test_dir.zip` containing the `test_dir` directory.

Extract ZIP File

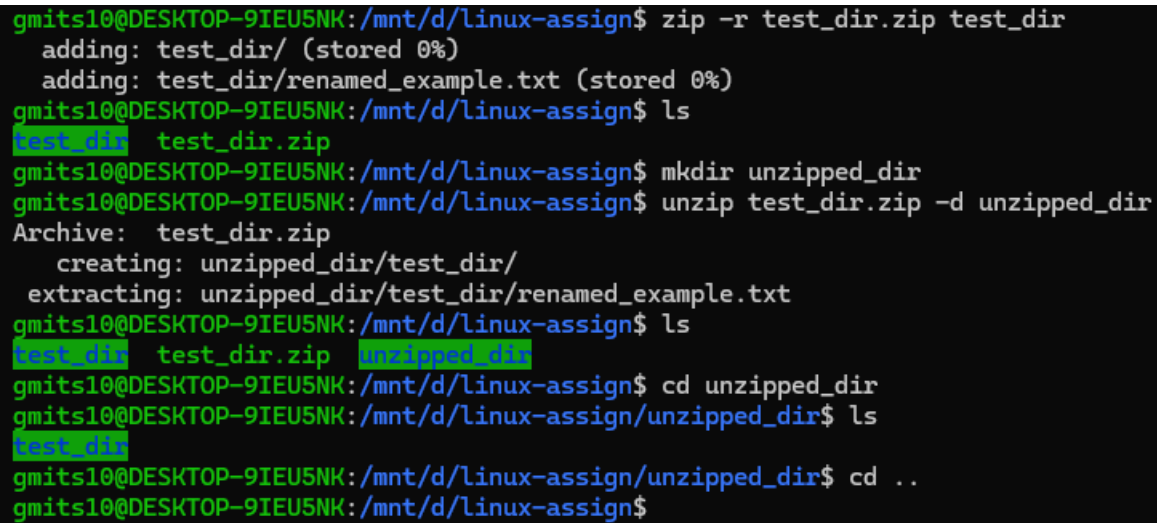
Command:

```
mkdir unzipped_dir
unzip test_dir.zip -d unzipped_dir
```

Explanation:

Extracts the contents of `test_dir.zip` into the `unzipped_dir` directory.

Screenshot:



```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ ls
test_dir  test_dir.zip
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ mkdir unzipped_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
  extracting: unzipped_dir/test_dir/renamed_example.txt
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ ls
test_dir  test_dir.zip  unzipped_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ cd unzipped_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/unzipped_dir$ ls
test_dir
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign/unzipped_dir$ cd ..
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$
```

5. Downloading Files

Download a File

Command:

```
wget https://example.com/sample.txt
```

Explanation:

Downloads the file from the specified URL into the current directory.

Note:

If the URL does not contain an actual file, use another valid download link.

Screenshot:

```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ wget https://raw.githubusercontent.com/github/gitignore/main/README.md
--2026-06-11 10:53:54-- https://raw.githubusercontent.com/github/gitignore/main/README.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ..
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.111.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5624 (5.5K) [text/plain]
Saving to: 'README.md'

README.md      100%[=====] 5.49K --KB/s  in 0.001s

2026-06-11 10:53:54 (4.58 MB/s) - 'README.md' saved [5624/5624]

gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ ls
README.md secure.txt test_dir.zip
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ head -n 8 README.md
# A collection of `.gitignore` templates

This is GitHub's collection of [`.gitignore`][man] file templates.
We use this list to populate the `.gitignore` template choosers available
in the GitHub.com interface when creating new repositories and files.

For more information about how `.gitignore` files work, and how to use them,
the following resources are a great place to start:
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$
```

6. Changing Permissions

Create a File

Command:

touch secure.txt

Make File Read-Only

Command:

chmod 444 secure.txt

Explanation:

Changes file permissions so everyone can read the file but no one can modify it.

Screen shot:

```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ cd
gmits10@DESKTOP-9IEU5NK:~$
mkdir permission-test
gmits10@DESKTOP-9IEU5NK:~$ ls
permission-test
gmits10@DESKTOP-9IEU5NK:~$ cd permission-test
gmits10@DESKTOP-9IEU5NK:~/permission-test$ touch secure.txt
gmits10@DESKTOP-9IEU5NK:~/permission-test$ chmod 444 secure.txt
gmits10@DESKTOP-9IEU5NK:~/permission-test$ ls -l secure.txt
-r--r--r-- 1 gmits10 gmits10 0 Jun 11 09:16 secure.txt
gmits10@DESKTOP-9IEU5NK:~/permission-test$ |
```

7. Working with Environment Variables

Create Environment Variable

Command:

```
export MY_VAR="Hello, Linux!"
```

Explanation:

Creates an environment variable named `MY_VAR`.

Verify Variable

Command:

```
echo $MY_VAR
```

Expected Output:

Hello, Linux!

Screenshot:

```
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ export MY_VAR="Hello, Linux!"
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ echo $MY_VAR
Hello, Linux!
gmits10@DESKTOP-9IEU5NK:/mnt/d/linux-assign$ |
```